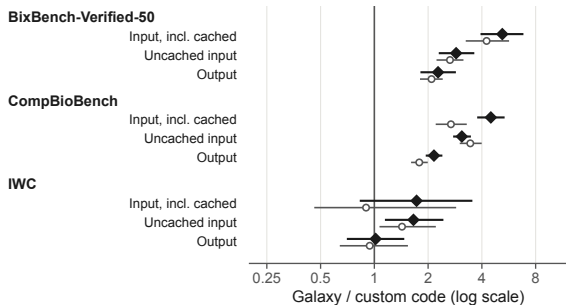


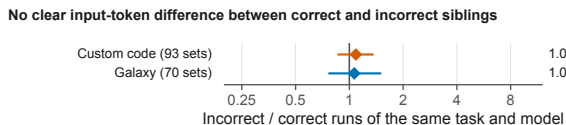
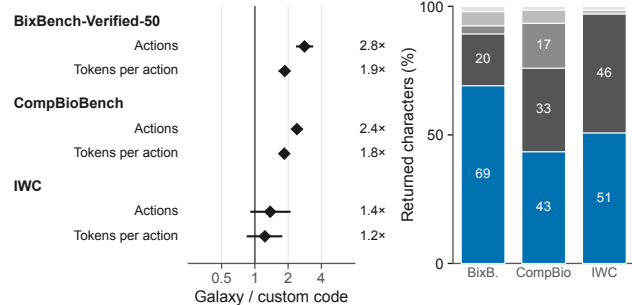
a Token use, Galaxy return, and custom code

Diamonds, typical paired task (geometric mean of task-model ratios; primary); circles, all tokens summed over the same tasks (aggregate)



b Where the extra input comes from

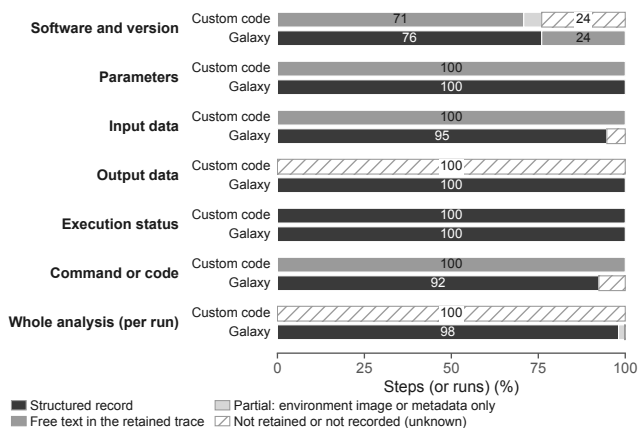
Left, Galaxy / custom code, typical paired task (all runs); right, the text Galaxy returned to the agent, by what the request was for



Tools inspected but not executed in the same run: 45–53% by benchmark.
Median share of input that was cached (earlier context reread): Galaxy 96%, custom code 93–96%.

c What the retained record holds for each analysis step

56,404 custom-code steps (shell commands labelled analysis) and 22,351 Galaxy jobs; whole analysis per run (1,920 runs per condition)



d One analysis step, recorded both ways

bix-45-q1, PhyKIT relative composition variability; GPT-5.6 Sol, replicate 1 (the case in Fig. 2d)

	Custom code: retained trace	Galaxy: job record
Software and version	PhyKIT 2.4.1 installed, then 2.0.3 downloaded and loaded through PYTHONPATH (in command text only)	phykit_metrics wrapper 0.2.0+galaxy0 (Tool Shed ID); library version not in the job record
Parameters	In the Python code of the command (file suffix, rounding)	Fields: metric = relative_composition_variability; mode = zip_archive; suffix = .mafft
Inputs and outputs	Input archives named in the command; output files not retained	1 input and 3 output datasets, linked by ID in the history
Status	Exit code 0 for every command	Job state "ok", exit code 0
Answer	Accepted	Rejected: the reference encodes the older PhyKIT value

The Galaxy record keeps fields a reviewer can query; the custom-code trace keeps the code, so the same facts must be read from it. Neither record was rerun to test reproducibility.